Lakshya JEE 2.0 (2025)

Physics

DPP: 10

Electric Charges and Fields

Q1 A cylinder of radius r and length L is placed in an uniform electric field E parallel to the axis of the cylinder. The total flux for the surface of the cylinder is given by
(A) zero
(B) $2\pi r^2 E$
(C) $\pi r^2 E$
(D) $\left(\pi r^2 + \pi L^2\right) E$

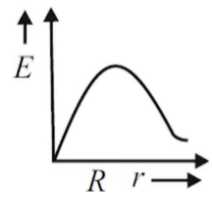
Q2 A cone of radius (R) and length (L) is placed in a uniform electrical field (E) parallel to the axis of the cone. The total flux for the surface of the cone is given by
(A) $2\pi R^2 E$
(B) $\pi R^2 E$
(C) $\frac{2\pi R^2}{E}$
(D) Zero

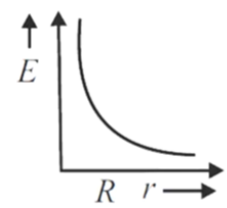
Q3 Consider a thin spherical shell of radius R consisting of uniform surface charge density σ. The electric field at a point of distance x from its centre and outside the shell is
(A) inversely proportional to σ
(B) directly proportional to x^2
(C) directly proportional to x
(D) inversely proportional to x^2

Q4 A long string with charge per unit length λ passes through a cube of length ℓ. The maximum flux through the cube will be
(A) $\frac{\lambda\ell}{\varepsilon_0}$
(B) $\frac{\sqrt{2}\lambda\ell}{\varepsilon_0}$
(C) $\frac{\sqrt{3}\lambda\ell}{\varepsilon_0}$
(D) $\frac{2\lambda\ell}{\varepsilon_0}$

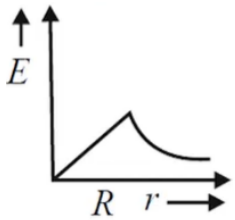
Q5 If ϕ_1 is the electric flux entering in a closed surface and ϕ_2 is the electric flux leaving the same closed surface, then the electric charge inside the surface will be
(A) $(\phi_1 + \phi_2)\,\varepsilon_0$
(B) $(\phi_2 - \phi_1)\,\varepsilon_0$
(C) $(\phi_1 + \phi_2)/\varepsilon_0$
(D) $(\phi_2 - \phi_1)/\varepsilon_0$

Q6 A uniform electric field $\vec{E} = a\hat{i} + b\hat{j}$, intersects a surface of area A. What is the flux through this area if the surface lies in the yz plane
(A) aA
(B) 0
(C) bA
(D) $A\sqrt{a^2 + b^2}$

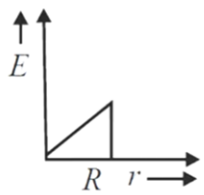
Q7 In a uniformly charged sphere of total charge Q and radius R, the electric field E is plotted as function of distance from the centre. The graph which would correspond to the above will be:
(A)


(B)


(C)



(D)



Q8 Let there be a spherically symmetric charge distribution with charge density varying as $\rho(r)=\rho_0\left(\frac{5}{4}-\frac{r}{R}\right)$ upto $r=R$, and $\rho(r)=0$ for $r>R$, where r is the distance from the origin. The electric field at a distance $r(r<R)$ from the origin is given by:

(A) $\frac{4\pi\rho_0 r}{3\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$

(B) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{5}{3}-\frac{r}{R}\right)$

(C) $\frac{4\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

(D) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{5}{4}-\frac{r}{R}\right)$

Q9 The region between two concentric spheres of P radii a and b , respectively has volume charge density $\rho=\frac{A}{r}$, where A is a constant and r is the distance from the centre. At the centre of the spheres is a point charge Q. The value of A such that the electric field in the region between the spheres will be constant is:

(A) $\frac{Q}{2\pi\left(b^2-a^2\right)}$

(B) $\frac{2Q}{\pi\left(a^2-b^2\right)}$

(C) $\frac{2Q}{\pi a^2}$

(D) $\frac{Q}{2\pi a^2}$

Q10 The electric field in a region is given $\vec{E}=\left(\frac{3}{5}E_0\hat{i}+\frac{4}{5}E_0\hat{j}\right)\frac{N}{C}$ The ratio of flux of reported field through the rectangular surface of area $0.2\ \mathrm{m}^2$ (parallel to $y-z$ plane) to that of the surface of area $0.3\ \mathrm{m}^2$ (parallel to $x-z$ plane) is a : b, where a =

[Here $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z-axes respectively]

Answer Key

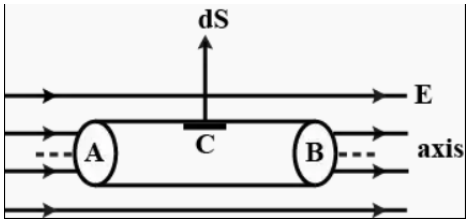
Q1	(A)	Q6	(A)
Q2	(D)	Q7	(C)
Q3	(D)	Q8	(B)
Q4	(C)	Q9	(D)
Q5	(B)	Q10	1

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

1. Let's understand this question with an image which is shown below:



2. Now the flux through the surface A is given as:

$\phi_A = E \times \pi R^2$

$\phi_B = -E \times \pi R^2$

3. Now the flux through the curved surface C is given as:

$C = \int E.\ ds$

$= \int E.\ ds\ \cos\ 90^0 = 0$

4. Hence the total flux through the cylinder is given by:

$= \phi_A + \phi_B + \phi = 0$

Video Solution:

Q2 Text Solution:

1. For the curved surfaces, at any point the vector E and the area vector are perpendicular to each other.

2. Hence the value of $\theta = 90^0$.

3. In this case, $\phi = E$.

4. Hence, $ds\ \cos\ 90^0 = 0$

Video Solution:

Q3 Text Solution:

As we know that,

The electric field at any point outside the shell is given by,

$E = \frac{kq}{R^2}$

where, k is contant, q is charge & R be the radius of the sphere.

or, $E = \frac{K\sigma(4\pi R^2)}{x^2}$

or, $E \propto \frac{1}{x^2}$ (σ and R are constant)

Hence, the electric field at a point of distance x from its centre and outside the shell is Inversely proportional to x^2.

Video Solution:

Q4 Text Solution:

It gives maximum flux only when the maximum portion of the rod lies in the cube.

So, it should be maximum in length in the cube when it is placed diagonally.

$\varphi_{max} = \frac{q_{enclosed}}{\varepsilon_0} = \frac{\sqrt{3}\lambda l}{\varepsilon_0}$

Hence, The maximum flux through the cube will be $\frac{\sqrt{3}\lambda l}{\varepsilon_0}$.

Video Solution:

Q5 Text Solution:

According to Gauss' theorem,

$\phi = \frac{q_{enclosed}}{\varepsilon_0}$

So, net flux = $\left(\phi_2 - \phi_1\right) = \frac{q_{enclosed}}{\varepsilon_0}$

$q_{enclosed} = (\phi_2 - \phi_1)\varepsilon_0$

Hence, the electric charge inside the surface will be $(\phi_2 - \phi_1)\varepsilon_0$.

Video Solution:

Q6 Text Solution:

As the surface lies in the yz plane,

So, the direction of the surface should be along the x-axis.

$\because$Area = $\mathrm{A}\hat{i}$

and, $\phi = \vec{E}\,\vec{A}$

$= (\mathrm{a}\hat{i} + b\hat{i})$

$= \mathrm{aA}$

Hence, If the surface lies in the yz plane, flux through this area is aA.

Video Solution:

Q7 Video Solution:

Q8 Video Solution:

Q9 Video Solution:

Q10 Video Solution:

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